

Beginner-Friendly Roadmap: Django To-Do App with CI/CD

Step 1: Setup Your Django Project

Create a Django project and a simple 'todos' app. Add a Todo model with title and is_done fields. Run migrations and add basic views and URLs to handle tasks.

Step 2: Run Locally

Run 'python manage.py runserver' and test your app locally.

Step 3: Create GitHub Repo

Push your code to GitHub. Use three branches: 'develop' for dev, 'staging' for staging, 'main' for production.

Step 4: Add Docker

Create a Dockerfile to containerize your Django app. Use docker-compose.yml to run the container easily. Run 'docker compose up --build' to test your app in Docker.

Step 5: Add CI (GitHub Actions)

Set up a GitHub Actions workflow (ci.yml) to run tests automatically when you push code. This ensures your app is working before deployment.

Step 6: Add CD (Continuous Deployment)

Set up GitHub Actions deploy workflows for each branch. - Push to 'develop' → deploys to Dev server - Push to 'staging' → deploys to Staging server - Push to 'main' → deploys to Production server

Step 7: Workflow

- Push code → GitHub runs tests (CI) - If tests pass, deploys automatically to the correct environment based on branch

Final Outcome

At the end you will have: - Django To-Do app (working locally) - Dockerized app - GitHub Actions for CI/CD - 3 environments: dev, staging, prod